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Artificial Intelligence (AI)

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Kick-Off Discussion

How do you use AI in your daily life?

How do you think AI change your life? In what ways?

Learning Objective

- Understand artificial intelligence
- Understand machine learning, neural networks, deep learning, natural language processing, computer vision systems
- Understand generative AI, intelligent agents, robots
- Understand the challenges of AI, responsible AI

What Is Artificial Intelligence

Artificial intelligence (AI) is a set of technologies that empowers computers to **learn**, **reason**, and **perform** a variety of advanced tasks in ways that used to require human intelligence, such as understanding language, analyzing data, and even providing helpful suggestions.

------ Google Cloud

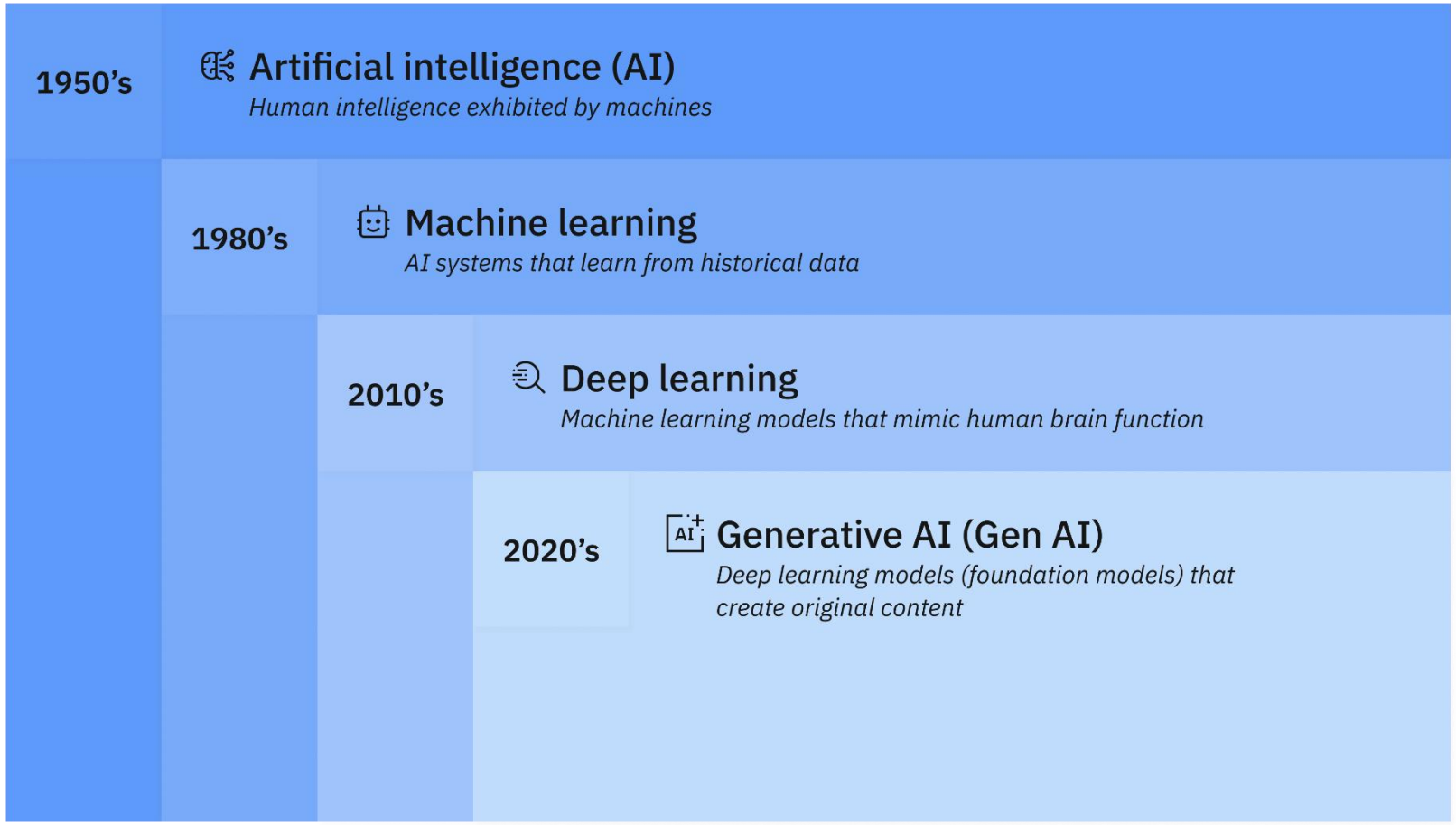
Artificial VS. Human Intelligence

- Speed & Capability
  - AI can process information faster and model some human abilities, but it cannot replicate full human cognition.
- Learning
  - AI: Learns through algorithms and large datasets, identifying patterns and making predictions.
  - Humans: Learn through genes, environment, experience, and social interaction, enabling flexible understanding.
- Adaptability
  - AI excels at specific tasks but needs retraining for new ones.
  - Humans quickly adapt to unfamiliar situations.

Artificial VS. Human Intelligence

- Language & Context
  - AI handles natural language processing, yet struggles with nuance, culture, and context that humans grasp naturally.
- Creativity & Intuition
  - AI generates ideas from data; humans innovate beyond existing patterns using intuition and imagination.
- Emotional Intelligence
  - AI can detect emotions but does not feel them; human emotions deeply shape judgment and relationships.

History of AI



Source: *"What is artificial intelligence (AI)?"* – IBM

Major Types of AI Technologies

- Machine learning
- Neural networks and deep learning
- Genetic algorithms
- Natural language processing
- Computer vision systems
- Robotics
- Intelligent agents
- Expert systems
- Fuzzy logic

Machine Learning

- From Rules to Learning
  - Early AI relied on rule-based systems (“if–then” logic) — rigid, narrow, and costly to maintain.
- What Is Machine Learning (ML)?
  - ML uses mathematical models that allow computers to learn from data and make statistical inferences with minimal human instruction.
- How It Works
  - Trains on large datasets to identify patterns and relationships.
  - Improves performance with more data and experience.

Machine Learning

Examples of ML Applications

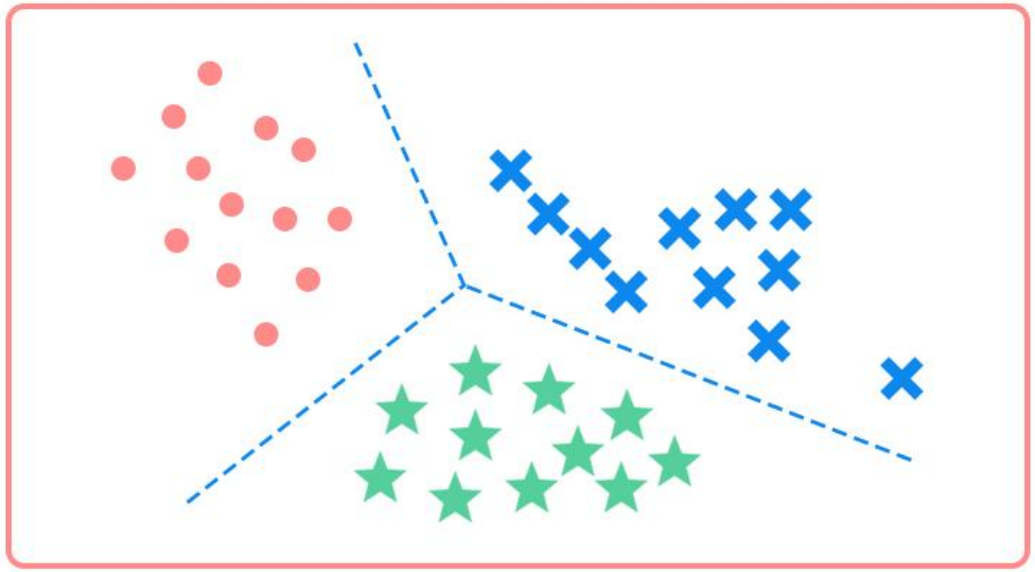
Application	Description
Natural language processing	Understanding and responding to human language
Image recognition	Software that can identify objects, places, people, writing, and actions in digital images
Predictive analytics	Predicting future events or trends based on historical data
Recommendation system	Recommending products or services to users based on their past behavior

Machine Learning – Supervised VS. Unsupervised

- Supervised Learning
  - The algorithm is trained on labeled data – examples with known inputs and outputs.
  - Learns by matching inputs to correct outcomes (e.g., identifying cat images).
  - Common in autonomous vehicles, fraud detection, and speech recognition.
  - Improves with more data and fine-tuning by programmers.
- Unsupervised Learning
  - Uses unlabeled data – no predefined outcomes.
  - Algorithm discovers hidden patterns or clusters on its own.
  - Example: grouping animal images by visual similarity without knowing species labels.
  - Applied in customer segmentation, anomaly detection, and market basket analysis

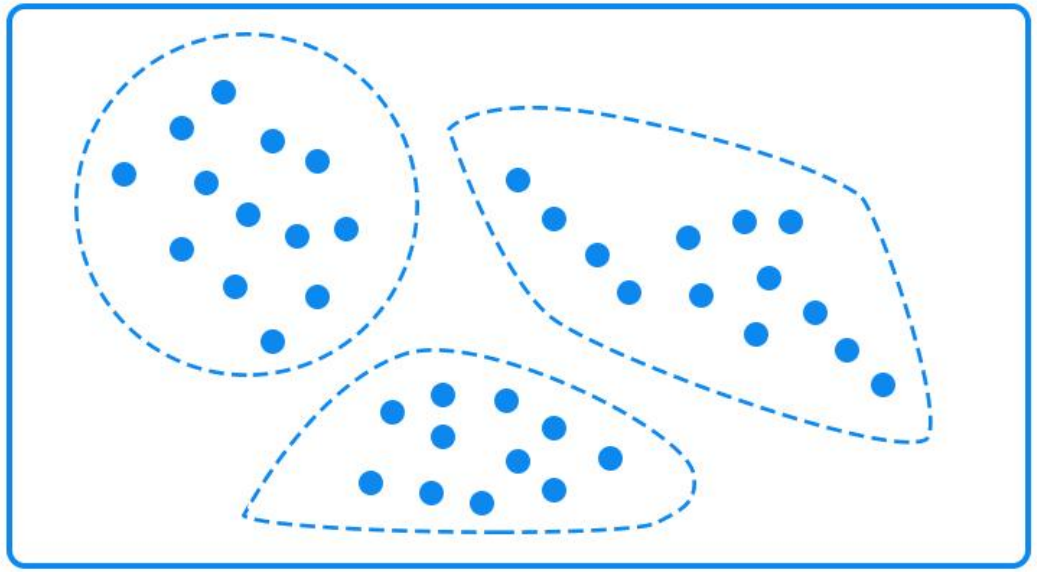
Machine Learning – Supervised VS. Unsupervised

Classification



Supervised Learning

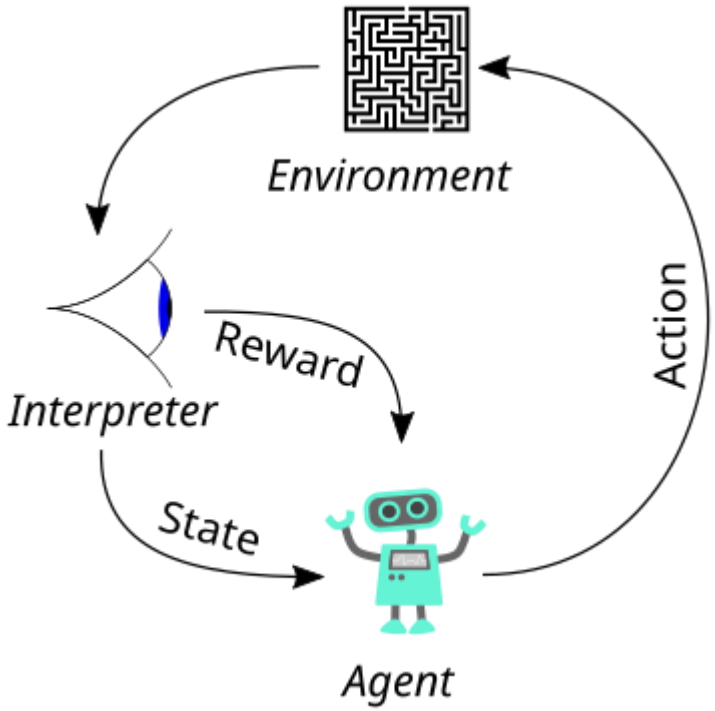
Clustering



Unsupervised Learning

Machine Learning – Reinforcement Learning

Reinforcement learning (RL) is a type of machine learning where an "agent" learns optimal behavior through interaction with its environment.



Recommendation Systems

- Streaming platforms like Netflix or YouTube use RL to continuously adjust recommendations based on how users interact (watch, skip, or like).
- The algorithm “rewards” itself when you watch a suggested video, reinforcing similar future suggestions.

Machine Learning

Examples of Unsupervised Learning

Application	Description
Customer segmentation	Groups customers by shared traits and behaviors to build profiles and target high-value audiences.
Medical imaging	Uses AI to detect, classify, and segment medical images for faster, more accurate diagnosis.
Anomaly detection	Finds unusual data patterns that may indicate fraud, defects, or system failures.
News aggregation	Collects and clusters online news articles in real time to organize evolving stories.

Quiz

Which of the following is an example of supervised learning?

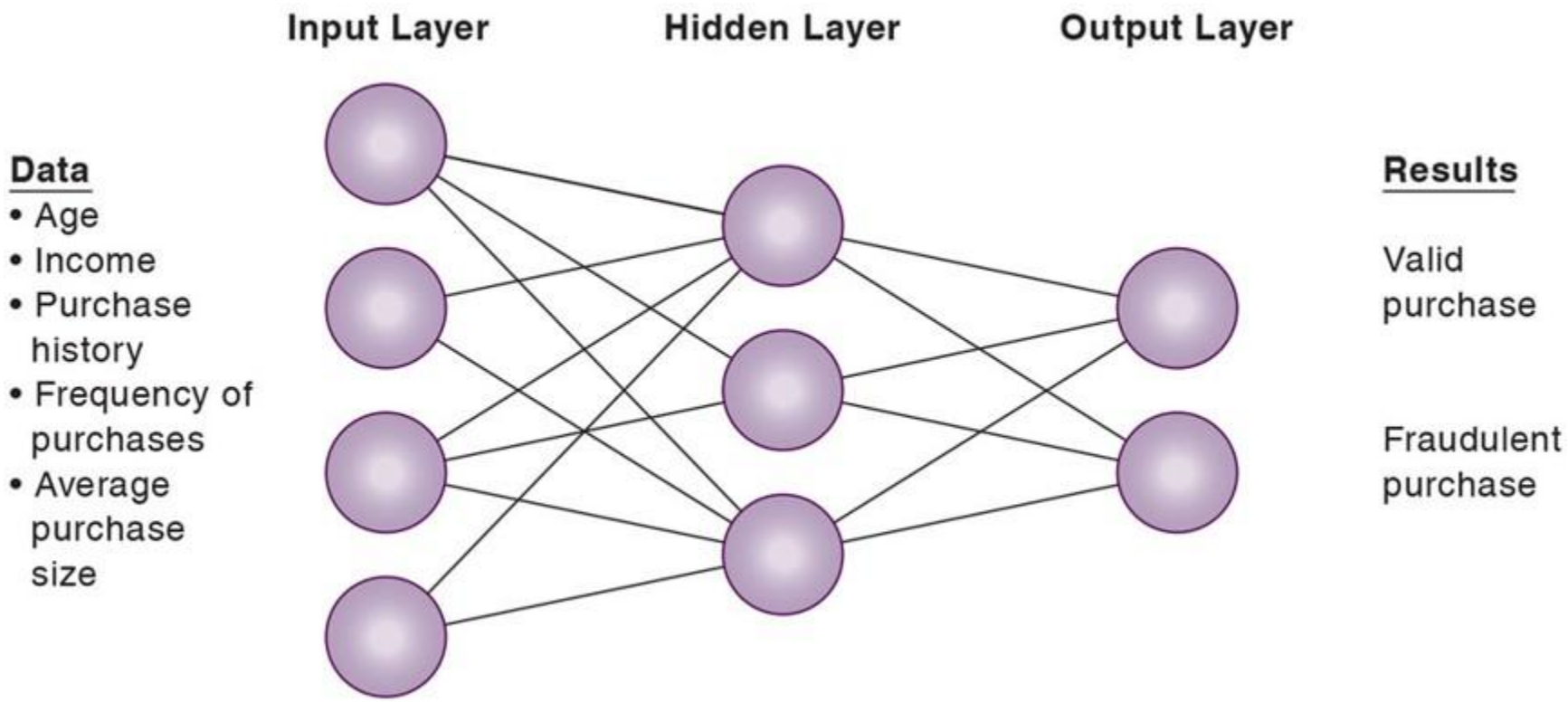
A. Grouping customers by shopping behavior
B. Training an algorithm to recognize spam emails using labeled examples
C. Detecting unusual credit card transactions without prior labels
D. Clustering news articles on similar topics

Neural Networks

- Identify complex patterns and relationships in massive datasets beyond human analysis.
- **Learn by iteration** – searching for patterns, building models, and refining through repeated correction.
- Trained by humans using labeled data so the network learns solutions by example.
- Applied in medicine, science, and business for classification, prediction, financial analysis, and optimization tasks.

Neural Networks

How a Neural Network Works

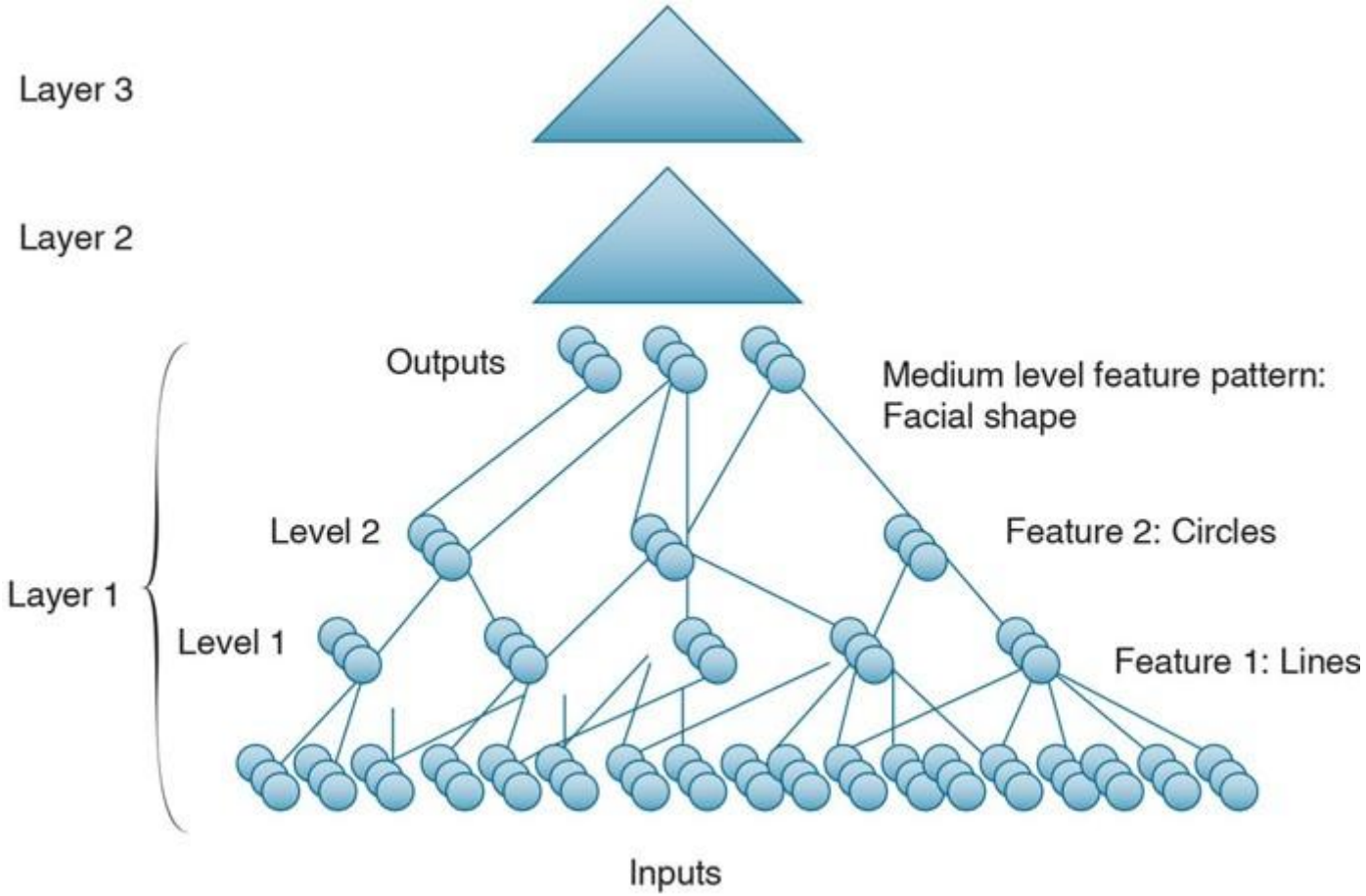


Deep Learning

- Deep learning neural network
  - More complex, with many layers of transformations of input data to produce target output
  - Used almost exclusively for pattern detection on unlabeled data (unsupervised learning)
  - Deep learning algorithms are powering many GenAI models

Deep Learning

A Deep Learning Neural Network



An unsupervised ML system that could identify cats and other objects without training

Examples of Neural Networks and Deep Learning Applications

Functionality	Inputs	Process	Outputs/Application
Computer vision	Millions of digital images, videos, or sensors	Recognize patterns in images and objects	Photo tagging; facial recognition; autonomous vehicles
Speech recognition	Digital soundtracks, voices	Recognize patterns and meaning in soundtracks and speech	Digital assistants; chatbots; help centers
Machine controls, diagnostics	Internet of Things: thousands of sensors	Identify operational status, patterns of failure	Preventive maintenance; quality control
Language translation	Millions of sentences in various languages	Identify patterns in multiple languages	Translate sentences from one language to another
Transaction analysis	Millions of loan applications, stock trades, phone calls	Identify patterns in financial and other transactions	Fraud control; theft of services; stock market predictions
Targeted online ads	Millions of browser histories	Identify clusters of consumers; preferences	Programmatic advertising

Natural Language Processing (NLP)

- Enables computers to understand and respond to human language — the way people naturally speak or write.
- Processes voice or text inputs to extract meaning, intent, or emotion.
- Common applications
  - Google Search and chatbots that understand queries
  - Spam filters that detect unwanted emails
  - Sentiment analysis of customer feedback
  - Call center systems that interpret and respond to spoken requests

💡 In business, NLP turns unstructured language data into insights for better decisions and customer engagement.

Quiz

Which of the follow statements about natural language processing is not true?

A. Natural language processing is typically based on machine learning.
B. Sentiment analysis is an example of natural language processing.
C. Natural language processing applies only to written text.
D. Many natural language processing algorithms are based on deep learning.

Examples of NLP Application

- Sentiment Analysis
  - Uses NLP to detect customer opinions and emotions about brands, products, or services.
  - Commonly applied to social media, reviews, and feedback to gauge public sentiment and improve customer experience.
- Conversational Agents
  - Power AI chatbots and virtual assistants (e.g., Siri, Alexa).
  - Understand and respond in natural language to user queries.
  - Widely used in customer service to automate responses and reduce support workload.

Computer Vision Systems

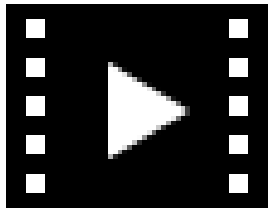
- Uses machine learning and deep neural networks to mimic human vision—analyzing and interpreting images or videos from the real world.
- Key applications
  - Autonomous vehicles (drones, self-driving cars)
  - Industrial inspection (e.g., quality control on production lines)
  - Military surveillance and defense
  - Robotics and automation tools
  - Medical imaging
  - Facial recognition
- Can systems still recognize faces when makeup, aging, or facial changes occur?

Quiz

Which of the following statements about computer visions systems is not true?

A. Computer vision systems only have a limited range of applications.
B. Computer vision systems can identify people.
C. Computer vision systems can recognize patterns in visual data.
D. Computer vision systems use machine learning.

Generative AI



- Definition
  - A type of machine learning model designed to create new content—not just analyze existing data.
- How it works
  - Learns patterns and structures from massive datasets (text, images, code, music, etc.).
  - Uses that learning to generate original outputs – text, images, audio, or video – that resemble human-created content.
- Prompt-based interaction
  - Starts with a prompt (e.g., a sentence, image, or melody).
  - AI models like ChatGPT, DALL·E, or Stable Diffusion produce responses – such as essays, visuals, or designs – based on that input.

Generative AI

- Foundation model
  - Underlying model that enables GenAI to work
  - Uses powerful neural networks designed to mimic the human brain; “learns” by finding patterns in existing data sets
- Large language models (LLM)
  - A foundation model trained on immense amounts of unstructured data covering many topics
- Transformer
  - A transformer is a key component of foundation models
    - For instance, GPT in ChatGPT stands for generative pre-trained transformer
  - Transformers are a type of neural network architecture that transforms or changes an input sequence into an output sequence

Intelligent Agents

- Software programs that operate autonomously, performing tasks without direct human control.
- Execute specific actions for users, business processes, or applications – often adapting based on data and context.
- Examples
  - Smart thermostats that adjust temperature automatically
  - Driverless cars that navigate in real time
  - Virtual assistants like Siri or Alexa
  - Chatbots for customer support
  - Spam filters that detect and block unwanted emails

Robotics

- Robotics
  - Deals with the design, construction, and operation of movable machines to substitute for humans
  - Control, sensory feedback, and information processing
- Robotic applications
  - Commercially available applications include the use of AIto enable robots to sense and respond to their environment
  - Most widespread use of robotic technology has been in manufacturing and logistics
- Types of robots
  - Service robots, medical robots, drones, autonomous cars

Challenges of AI

- Hallucination
  - AI systems may generate false or misleading information that appears credible
- Lack of Transparency
  - Many AI models function as “black boxes,” making it difficult to understand how they reach conclusions.
- AI Bias
  - Algorithms can inherit or amplify biases present in training data, leading to unfair or discriminatory outcomes.
- Cybersecurity Problems
  - AI can both create and defend against cyber threats; malicious actors may use it to automate attacks or generate phishing content.

Challenges of AI

- Job Loss and Inequality
  - Automation may displace workers and widen the gap between high-skill and low-skill jobs, raising social and ethical concerns.
- Environmental Impact
  - Training large AI models consumes massive computing power and energy, contributing to carbon emissions and electronic waste.
- Misinformation and Disinformation
  - Generative AI can create convincing fake text, images, or videos, making it easier to spread false or harmful content.
- Loss of Privacy
  - AI systems often rely on vast amounts of personal data, increasing risks of surveillance, data misuse, and unauthorized profiling.

Responsible AI

- Responsible AI refers to the design, development, and use of AI systems in ways that are ethical, transparent, fair, safe, and aligned with human values, ensuring AI benefits individuals, organizations, and society.
- Key Dimensions
  - Fairness & bias mitigation, transparency & explainability, accountability, privacy & data protection, security & safety, human-centered design

AI Ethics & Governance Course

Quiz

A financial institution adopts an AI system for loan approval. Later, it’s discovered the model rejects more applications from certain demographic groups.

What AI challenge does this illustrate, and how might the company address it?

Prompt Engineering

Prompt Engineering 101 - Crash Course & Tips

Summary

- Artificial Intelligence (AI)
- Machine Learning & Deep Learning
  - How ML models find patterns in data; how neural networks and deep learning enable advanced capabilities.
- Key AI Technologies
  - NLP – systems that understand and generate human language
  - Computer Vision – systems that interpret images and video
  - Generative AI – models that create new text, images, and designs
  - Intelligent Agents & Robots – autonomous systems acting on data and goals
- Challenges & Responsible AI
  - Hallucinations, bias, transparency, cybersecurity, privacy, environmental impact, and social implications (job loss & inequality).Importance of governance, ethics, and human oversight.